

COMMENTARY

# Leveraging Data Science to Advance Implementation Science: The Case of School Mental Health

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**Citation:** Carlson C, White SW, Hudnall M, Lochman J, Laird R. Leveraging data science to advance implementation science: the case of school mental health. *J Sch Health*. 2023; 93: 1045-1048. DOI: 10.1111/josh.13385

Received on September 27, 2022

Accepted on July 17, 2023

Data science (DS) is the formalized discipline of managing the full lifecycle of data across areas of scientific inquiry. Although the field of DS has existed for almost 50 years, the evolution of the tools, frameworks, and processing power needed to provide meaningful advancements have only recently come to fruition. DS has been integrated into multiple industries and disciplines to optimize outcomes, increase efficiency, and maximize human and financial resources. Implementation and health services research has yet to harness DS to facilitate sustained uptake of evidence-based interventions (EBI). In this commentary, we describe how implementation and services researchers and policy-makers might leverage DS to advance school health, with a focus on school mental health (SMH). Implementation research on SMH examines factors affecting the adoption of EBIs appropriate for school contexts, training and professional development of school personnel, maintaining intervention integrity during school-based delivery, and EBI sustainment in schools. We propose three DS approaches that may support the advancement of SMH implementation, and school health research more broadly: data standardization and linkage, data analytics, and data visualizations. Finally, we suggest ideas for potential SMH implementation research and practice utilizing these DS approaches to optimize service delivery in schools.

At the intersection of statistics and computer science, the discipline of DS supports data-driven decision-making across areas of scientific inquiry. Defined as “a set of fundamental principles that support and guide the principled extraction of information and knowledge from data.”<sup>1</sup> The growth of DS has been fueled by the emergence of big data,<sup>2</sup> characterized as data that is too large for a single system or combination of systems to process within a reasonable timeframe. DS-based approaches such as machine learning and advanced algorithms can incorporate human expertise and data to generate artificial intelligence (AI) models. These models can be used to make complex decisions based on inputs and execute actions accordingly. Numerous examples from diverse fields—such as supply chain management, e-commerce, and health care—attest to the power of leveraging DS to help solve challenges fundamental to their fields. The application of DS to the optimization of mental health care and service delivery has been previously suggested.<sup>3</sup> IS has yet to harness DS to facilitate implementation of EBI for mental health in schools.

Many psychiatric disorders and problem behaviors (e.g., suicide, violence) in youth are treatable, and sometimes preventable, through EBI.<sup>4</sup> Unfortunately, most youth do not receive any services much less EBI,<sup>5,6</sup> which are needed now more than ever in the era of COVID-19.<sup>7</sup> Children are more likely to receive

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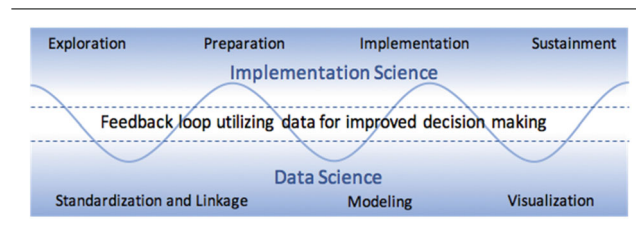
This work was supported by the National Institute of Mental Health (K01 MH110651 to Catherine Carlson).

services at school than through any other delivery sector.<sup>8</sup> Enthusiasm for the effectiveness of EBIs in schools is tempered by concerns about the extent to which programs are implemented as intended and able to be sustained over time. School contexts present unique challenges in mental health implementation and dissemination including educational timelines, diverse professional characteristics, organizational constraints, legacy computer systems, and state/federal policies, which change over time and regime. Even when schools are committed to implementing EBIs, challenges arise that might be best understood via systematic consideration of data.<sup>9</sup> Mental health services delivered in most schools are neither consistently evidence-based nor rigorously implemented, and often there is no system in place for ongoing evaluation of the data to determine impact, leading to suboptimal clinical and educational outcomes, and contributing to a resource siphon within an already-strained system.

The latest implementation literature for SMH highlights specific areas of needed attention, including how to choose and adapt interventions to fit school contexts, training and professional development of school personnel in EBIs, maintaining intervention integrity in school settings, and sustainment of EBIs in schools over time.<sup>10</sup> Enhanced use of data can support the next wave of implementation research in these areas, such as using data to inform decisions on intervention selection, monitoring implementation processes through real-time data collection and use, and building systems to track student and school outcomes. DS strategies could also support the use of data across multilevel school interventions for the convergence strategies to optimize student and implementation outcomes.<sup>11</sup>

Measurement and data are ineradicable from education (e.g., attendance tracking, standardized testing). Yet, most schools, districts, and states struggle to effectively utilize data, especially for mental health, in part due to antiquated data management systems that do not produce user-friendly tools for interacting with the data, adeptly manage the volume and variability of data, or permit school personnel less familiar with data management to utilize the data for immediate purposes.<sup>9,12</sup> Most school systems already implement student information systems (SIS) that track many data points, including grades, assignments, attendance, teacher comments, disciplinary actions, safety notifications, emergency alerts, and other elements. As of 2021, PowerSchool, the most widely used SIS, was leveraged by over 45 million students across 13,000 schools and represents 90 of the top 100 school districts.<sup>13</sup> Given the large amounts of data already available to schools and districts, using these data to provide informed decision-making and intervention programs is the logical next step in SIS data collection. Herein, we propose three specific

Figure 1. Convergence of IS and DS



ways in which DS might help overcome challenges associated with the effective use of data for SMH implementation, with real-world examples. Couched in the EPIS framework,<sup>14</sup> Figure 1 depicts how DS and IS can converge through ongoing feedback loops utilizing data standardization, analytics, and visualizations throughout phases of implementation.

Many factors inhibit the use of data for SMH decision-making. Foremost among these is that most data systems used now were developed only to collect data, not to be reciprocal, or assist school personnel with the use of that data. Further, schools are often overburdened with the volume of data and its unwieldy presentation, and lack staffing and resources needed to effectively use student mental health data for decision-making. Data standardization and harmonization (i.e., using identical assessments, collected on the same time frames, across sites and projects) could render direct comparisons across schools, systems, and EBIs more attainable, to aid data-based decision making. The linkage of big data sources, such as education, hospital, and mental health services, has been suggested as a means to inform and evaluate service delivery. Data standardization in health care, for example, has helped to bridge gaps in health records transfers using Fast Health care Interoperability Resources (FHIR) for patient data repositories to help combat COVID and ease the overall transfer of records from system to system.<sup>15</sup> Such linking in schools could occur through use of natural language processing software (a branch of AI) to extract data from clinical progress notes or correspondence which is then linked to a school, mental health, and hospital database.<sup>16</sup> Practically, steps such as linking student ID numbers across systems and including fixed response categories (e.g., student characteristics, current symptoms, interventions used) as tags in case notes would support data use and efficiency. Linking data from across multiple, complex systems from school-level (achievement, disciplinary, climate, etc.), providers (training, number, experience, etc.), child (medical records), or state departments of education could be used to best predict potential school-intervention fit for decision makers charged with identifying mental health EBIs for their context. Importantly, data standardization does not preclude

individual school systems from using assessment tools to meet unique needs or from adopting standardized measures that combine idiographic and nomothetic approaches.

Schools are inherently complex, multi-tiered systems. Many schools, for example, do not employ their own providers; rather, services are delivered by providers from external mental health clinicians who work in schools under contractual arrangements. School-based mental health care involves interconnected organizational units (i.e., schools, families, and community agencies) and requires the linkage and analysis of multiple streams of data from various sources (e.g., teacher, parent). DS can contribute to the analysis of complex data across multiple levels and sources. Consolidated data repositories and the associated data analytics leveraged upon them have been shown to help “connect the dots” between disparate data sources to assist in other fields (such as combating substance use disorders, unearth connections in disease contributory factors, provide wholistic views of patients in both mental and physical health settings) and provide a data-informed basis for overall decision making.<sup>15,17-22</sup> School data and the opportunity to leverage these data to improve student well-being presents a significant untapped resource for improving student outcomes with interventions. While IS has increasingly identified relevant factors influencing implementation, less is known about the relationships among these factors. As has been used in clinical research (i.e., precision medicine), DS analytics (such as machine learning algorithms) can be used during an exploration phase to integrate data across complex implementation levels into a predictive tool to aid in decision-making on what intervention or strategy will most likely result in successful implementation and sustainment. Such modeling can also be useful in helping to answer research questions when other approaches, such as controlled trials, are not feasible (as is often the case in implementation research) or acceptable to the school community. During implementation, ongoing and real-time digitized collection of intervention data (i.e., through mobile applications) could use AI algorithms to inform providers and/or supervisors when an EBI fidelity is at risk of falling below predetermined acceptable levels. Such information can then prompt renewed attention to or inquiry into reasons for failing intervention integrity.

Another area in which DS can support the advancement of IS within SMH is the packaging of data for consumers. DS ensures the usability of data for practitioners and decision-makers, sometimes referred to as citizen data scientists. In the context of SMH, decision-makers, such as school and system-wide administrators, are often busy, non-data experts weighing multiple priorities at once. Simplified and interactive data visualization with real-time data inputs can help

with decision-making related to SMH implementation, especially given the identified challenges of improving educators capacity to use and understand data.<sup>12</sup> For example, the use of dashboards—a DS tool commonly used by corporate analysts—may facilitate the use of data for decision-making and implementation. Development of user-friendly dashboards that require minimal data literacy for providers and decision-makers to continue monitoring can assist schools with ongoing sustainment of EBI over time. Through individualized queries, such dashboards can convey the ongoing success (or challenges) of specific implementation and clinical outcomes. Examples of dashboards in schools already exist that have the opportunity for wider use. The Early Identification System is being used in six Midwestern school districts to identify students at risk of mental health concerns through the use of data analytics and dashboards.<sup>23</sup> Funded through a local tax and intended to not have reoccurring costs, this universal screener is administered and scored online and provides instantaneous, consumable data for existing school personnel via a dashboard report. Another example, PBISApps (<https://www.pbisapps.org/>), has developed a suite of apps that include dashboards in support of all three Tiers of the PBIS approach, from system-wide to individual student data.

DS provides many tools, of which we have highlighted three, that can support the complex and ever-changing landscape of health services research broadly, and SMH specifically. While perceived hurdles exist, which we have outlined, examples of success also exist that can be implemented within real-world resource constraints and by everyday school and mental health personnel. The field can be further advanced through in-service workshops for school personnel on existing approaches, integrating DS fundamentals into education and mental health training programs, as well as increased multidisciplinary collaborations involving data scientists within school health implementation research. DS holds the potential to improve our use of increasingly available and complex data to further the goals of implementation research and practice. Perhaps the central barrier to the deployment of DS in schools is also the fundamental reason *why* DS can be so critical to parsimonious and efficient decision-making: schools operate as fairly autonomous systems and often rely on cumbersome, sometimes even antiquated, systems for data management and student tracking.

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